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Review

Handover schemes and algorithms of high-speed mobile environment: A survey

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ABSTRACT

With lower energy consumption, less environmental pollution, larger capacity, and higher safety, the high-speed railway is playing an important role in mass transportations. The development of high-speed railways makes people's life more and more convenient. Meanwhile, it puts forward much higher demands on high-speed railway communications. Therefore, new solutions are desired to keep up with this unprecedented growth. However, communications in high-speed environment have particular characteristics such as fast and frequent handover, high penetration loss, and high Doppler frequency shift. Current researches mainly focus on providing broadband wireless network access for high-speed mobile terminals. This article presents an overview of the latest technical developments on handover schemes and handover algorithms targeting high-speed mobile scenarios. In addition, this paper suggests the most appropriate techniques which can efficiently communicate in high-speed mobile scenarios. Finally, we discuss future trends of handover researches in high-speed mobile environments.

Currently, Cyber-Physical System (CPS) is emerging on the basis of environmental perception and deep integration of computing, communications, and control (3Cs) technologies. CPS is a device-networking system which is controllable, trusted, and scalable. Through interactions and integrations between the calculation process and the physical process, the system can increase or extend its functionalities. Therefore, it can monitor or control physical entities via much more safe, reliable, efficient and real-time ways [1–7]. China Train Control System Level 3 (CTCS-3) is designed for high-speed mobile environments. As one of the CPS subsystems, CTCS-3 shall meet the CPS characteristics which include reliability, heterogeneous, autonomous, real-time performance, green energy, large capacity, and safety. The high-speed railway (HSR) is playing an important role in mass transportations not only in China, but also in other parts of the world. The development of high-speed railways makes people' life and work more and more convenient. Meanwhile, this development puts forward various higher service requirements for high-speed (i.e. above 300 km/h) users.

For the purpose of ensuring system reliability and real-time performance, underlying technologies must be reliable and real-time in the first place. In addition, the reliability and real-time performance of communication technologies are mainly determined by the handover performances. Therefore, in order to provide Quality of Service (QoS) for high-speed users, the handover is a key element in wireless cellular networks. In addition, it is also a main carrier of the CPS performance metrics (e.g. delay). As illustrated in [8–10], according to future consumers' requirements, broadband high-speed mobile communication is an unavoidable trend. Innovative technology solutions are desired to keep up with this unprecedented demand growth.

The scope of this paper is to provide a state-of-the-art overview of different types of handover schemes and handover algorithms on broadband high-speed mobile scenarios, to classify them and to discuss them accordingly. The rest of this paper is organized as follows: Section 1 analyzes the problems caused by high-speed mobility. Basic handover concepts are presented in Section 2. Various handover schemes are discussed in Section 3. Section 4 summarizes different handover algorithms. Besides, comparisons are also made in Sections 3 and 4, respectively. Finally, Section 5 concludes this paper.

1. Effects of high-speed mobility

The wireless connection between the high-speed train (HST) and the ground network is the bottleneck of improving the QoS of train passengers [11,12]. The wireless access problems in high-speed mobile scenarios are as follows: (1) Frequent handover, which causes frequent cell re-selection, call quality decline, and

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data service unavailability; (2) Doppler frequency shift and fast fading, which causes intolerably low communication connection rate, difficulty in handover initiation, and numerous handover failures; (3) Large car body loss and multipath loss, which makes the intra-train to be a signal weak field. In this section, these effects will be discussed in detail.

1.1. Fast handover

An important factor which affects the broadband wireless access for HSTs is the fast and frequent handover. The frequent handover is caused by the high-speed mobility of mobile terminals. For a given cell size, the higher of the HST speed, the shorter the time it spends to pass through the overlapping region. When the minimum handover processing delay is larger than the time interval of the HST passing through the overlapping area, the handover process fails to complete, resulting in call drops. For example, the cell of a Remote Antenna Unit (RAU) typically has a diameter of 100 m. Then for the HST at a speed of 100 m/s, handovers will occur once every second. So, traditional handover times of 0.1–1 s are definitely unacceptable [10]. Frequent handover also brings the problem of packet loss and packet reordering [13]. Therefore, a main difficulty that the high-speed railway broadband wireless access faces is how to achieve fast handovers.

In order to meet the requirement of fast handovers, on the one hand, increasing cell radius can reduce the handover rate (i.e. the number of handover events per unit time). However, according to [14], expanding cell coverage will sacrifice part of the system capacity. In the special HSR scenario, improving handover mechanisms and optimizing handover decision conditions and procedures will help to reduce the handover time. In this way, stability and reliability of the network performance are guaranteed. On the other hand, reducing the cell radius (e.g. 100–500 m) and applying both the optical switching and the millimeter wave technologies can achieve fast handovers. Millimeter wave, e.g. 60 GHz RF signal, has a good line of sight (LOS) characteristic in free space, and its transmission distance is around 100 m. All these properties will help eliminate the multipath effect. However, this solution increases the number of handover events. The handover rate will be too high in a high-speed mobile environment.

1.2. Doppler frequency shift

The signal frequency shift at the receiver due to the movement is known as the Doppler frequency shift. During the train traveling, Doppler frequency shift can be calculated by $f_d = \frac{v}{c} f_c \cos\theta$, where v is the train speed, c is the speed of the electromagnetic wave, f_c is the central carrier frequency, and θ is the angle formed by the signal and the train direction (Fig. 1). The center frequency of the wireless signal (e.g. uplink f_0, downlink f_1) shifts dramatically because of the Doppler Effect in high-speed mobile scenarios. When the train is crossing the handover zone, the downlink frequency shift is jumping from $+f_d$ to $-f_d$, with a total change of $2f_d$. For instance, a central carrier frequency of TD-SCDMA (Time Division-Synchronous Code Division Multiple Access) is 2025 MHz, and if the train speed is 350 km/h, the Doppler frequency shift at the receiver is about ±656 Hz, which will destroy the orthogonality of subcarriers in the OFDM (Orthogonal Frequency Division Multiplexing) system. This will induce Inter Channel Interference (ICI), which further deteriorates wireless channels, causes significant impacts on bit error rate (BER), and increases the synchronization difficulty.

In order to reduce the Doppler frequency shift, base station (BS) sites should be deployed far away from the tracks in the initial network planning. In this way, the angle θ of the multipath waves arrived at the receiver is large enough to mitigate the Doppler frequency shift. However, in practical network deployment, for the penetration ability enhancement of the wireless signal, there is usually a short distance between the base station and the track. Therefore, another technical difficulty of the high-speed railway broadband wireless access network is the Doppler frequency shift compensation.

1.3. Penetration loss

Another major factor in the high-speed railway broadband wireless access system is the penetration loss. The signal suffers this loss when it goes through train compartments. Most kinds of high-speed trains are metal bodies with large windows of a single layer or multilayer glasses. The entire train compartment is like a Faraday cage which increases the penetration loss. For example, penetration loss of the CRH (China Railways High-speed) series HST is 10 dB larger than ordinary trains. In addition, with the increase of frequency, signals that go through the compartments will suffer greater attenuation. Meanwhile, the smaller the wireless signal incidence is, the higher the loss it bears.

In order to overcome the penetration loss, various methods such as reinforcing the trackside base station transmitter power, upgrading the base station receiver sensitivity, and enhancing the transmission power of the user terminal, can be applied. However, none of these methods corresponds to the Green Communication concept. A much different approach is using the two-hop model [10] which introduces a dedicated multi-band and multimode base station (acts as a relay station) for the HST. The first hop is from the BS to the HST, and the second hop is from the HST to the user terminals (Fig. 2). This model architecture will be

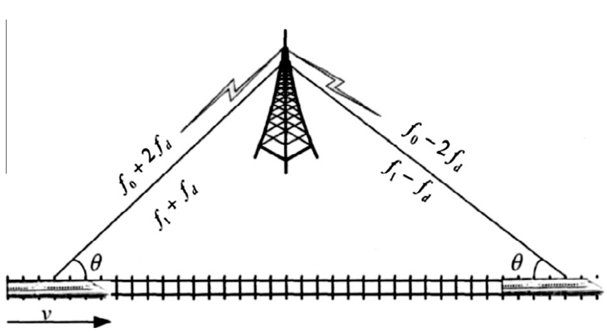

Fig. 1. Doppler Effect on high-speed rail.

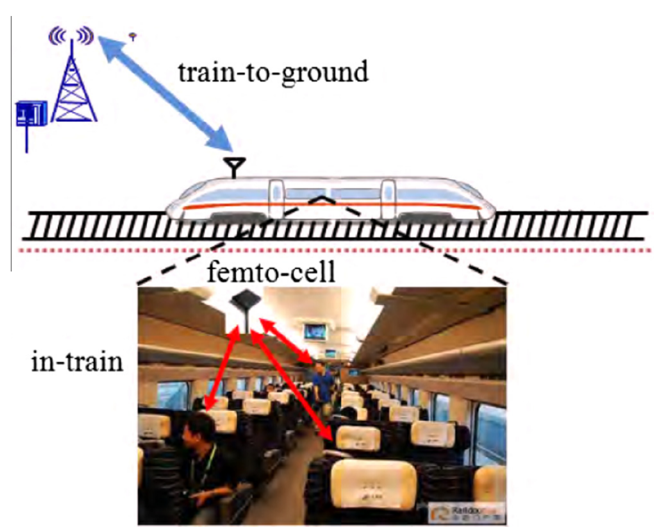

Fig. 2. Two-hop architecture.

frequently mentioned in the rest of this paper. In this model, wireless signals of various standards are aggregated and fed into the compartment through the antennas erected on the HST roof top, and then reach the user terminals via distributed antenna systems within the train compartments. By applying the two-hop model, penetration losses can be avoided. Furthermore, the HST can be treated as an individual terminal that directly communicates with the BS.

2. Handover

Handover (HO) is the mechanism that transfers an ongoing call from one cell to another as a user moves through the coverage area of a cellular system [15]. Handover is a key element in a wireless network to provide continuous connections and user perceived QoS. For the sake of readers' convenience, this Section recalled some main concepts related to handovers. Some performance metrics of handover are presented in this paper. Detailed information about handover process is illustrated in [16].

2.1. Handover phases

A classical handover procedure mainly consists of the following steps:

Initiation: The mobile station (MS) measures channel qualities, such as Received Signal Strength (RSS), Signal to Interference Ratio (SIR), distance, and BER, etc., of different BSs. Then MS sends the report back to the source BS.

Network selection: The source BS forwards the measurement report to the control unit (CU). Taking into account the resource availability and the network load, the CU decides which BS is the best to maintain the connection. The handover decision-making process may be centralized or decentralized based on where the decision is made (see Section 2.2). If a handover is required, then the CU informs the target BS to prepare a connection, and lets the source BS send handover commands to the MS.

Execution: The MS starts to set up a connection with the target BS using additional resources by operations such as authentication, synchronization, network reconfiguration, etc. Finally, the MS sends a signal to the target BS indicating the completion of the handover.

It can be seen that these procedures involve many wireless and backbone signaling and data exchanges. It may also take a certain time to complete the handover. In Section 4, some reviews on the handover decision-making algorithms are provided.

2.2. Types of handovers

Handovers can be distinguished into horizontal handovers and vertical handoffs according to the access technologies involved in. On the one hand, horizontal handovers occur within a single type of network interface which involves only one wireless access technology. In a cellular network, there are two types of horizontal handovers: intra-cell handovers and inter-cell handovers. Intra-cell handovers only change radio channels within a cell in order to minimize the inter-channel interference. Inter-cell handovers change all the connections from the source BS to the target BS when the MS moves between two adjacent cells. On the other hand, vertical handoffs change the connection between a variety of different network interfaces which typically have different wireless access technologies, such as 3GPP (3rd Generation Partnership Project), 802.11, and 802.16, etc. The authors in [17] design a 3G (Third Generation in Mobile Communication System) and Wi-Fi complementary scheme in the vehicular movement scenario, providing a reference for high-speed mobile scenario handover designs. Vertical handoffs have many advantages, e.g. the network features and user demands are combined together when the user switches between different subnets. These technologies not only retain the traditional handover functions, but also ensure the users to enjoy the best services anytime and anywhere. After the handoff is performed, the user does not even perceive that he or she had entered a more ideal network to enjoy better services. From the network perspective, vertical handoffs can efficiently utilize the bandwidths and balance the network loads. For specific overviews of vertical handoffs, please refer to [18,19].

Based on where the handover decision is taken and controlled, handovers can be classified into three types: Mobile-executed Handover (MCHO), Network-executed Handover (NCHO), and Mobile-assisted Handover (MAHO). In the MCHO, the MS is continuously monitoring the signals and the interference levels of different BSs. A handover is initiated by the MS if the signal strength of the serving BS is below a threshold. The MCHO is typically used in the digital cordless telephone system. The BSs measure the signal from the MS in the NCHO. The network initiates handover procedures when the criteria of SIR and BER are satisfied. The NCHO is mainly used in the first-generation communication system. For the MAHO, the network requests the MS to measure signals from the BSs. Then the network makes decision based on the reports from the MS. The MAHO is mainly used in the second-generation (2G) communication system.

The typical handover type in traditional cellular mobile communication systems, like GSM (Global System for Mobile Communications), is hard handover, i.e. the implementation of the "break before make" process. In this procedure, only one traffic channel is activated (Fig. 3). There is a short disconnection on the radio channel in order to achieve the conversion of carrier frequencies. Therefore, hard handover scheme is more likely to cause call drops. An enhanced version of GSM named GSM-R (GSM for Railway) is applied in Europe and China for railway communications. However, in the GSM-R system, all users within a single train simultaneously initiate handovers to the same BS. Its call dropping rate and congestion could be enormous [20]. Inevitably, this system suffers low data rate and unreliable link connection. So this system does not meet the broadband communication requirement in the high-speed mobile scenario.

With the emergence of 3G, soft handover is widely used between cells that use the same carrier frequency (Fig. 3). In the soft handover procedures, the user terminal maintains wireless communication links with the source BS and the target BS (i.e. more than one BS) at the same time. Then the terminal disconnects the link with the source BS after handover completion, i.e. the implementation of the "make before break" process. In this way, the communication service quality is greatly improved. However, soft handovers can only be applied to the same carrier frequency. In [21,22], the authors find that soft handovers in conjunction with beamforming technique could significantly enhance system throughput, reduce handover delay, and decrease BER in the cellular CDMA (Code Division Multiple Access) system. Softer handover, another handover type, is performed between the sectors within the same cell or between the sectors of different cells. This procedure is controlled by the BSs directly, so there is no need to route the user data to the mobile switching center (MSC). There is a novel handover type called relay handover which is used in the TD-SCDMA system based on smart antenna technology [23]. Relay handovers take the advantage of precise positioning technology, i.e. the BS uses user terminal's distance and direction parameters as auxiliary information to determine whether to handover the user terminal. This strategy can improve the handover success probability, reduce the handover rate, save the channel resources, simplify the signaling process, and lower the system load. In addition, relay handovers can be performed between cells that are

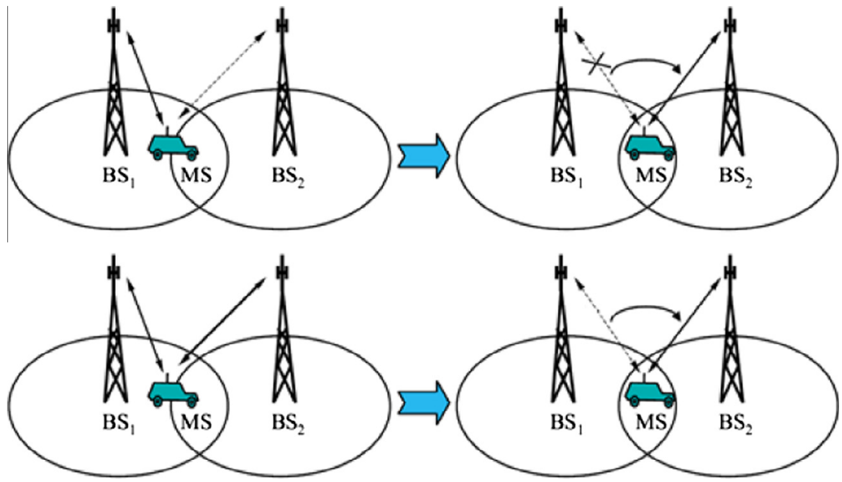

Fig. 3. Hard and soft handovers.

using different frequencies. The relay handover, to some degree, enlightens us to design fast handover strategies for HSR communications.

Vertical handoff is a common phenomenon in B3G (Beyond Third Generation in Mobile Communication System). Since different types of handovers can be employed and integrated in B3G, handoffs are more complex. The fundamental change in the core network of the Long Term Evolution (LTE) system is mainly driven by the evolution which is based on all-IP services [24]. Under the flat network notion, handovers are performed directly between eNBs. The handover procedures in LTE system are similar as previously described. The major difference is that it could handoff to a 2G or 3G network when the LTE network is not available. Besides, various new techniques are applied to improve the performances and the procedures of handovers. For instance, 3GPP LTE recommends a layer-2 mechanism called data bi-casting to reduce cell switching interruption time and support QoS for real-time services. The idea of bi-casting is to make the data available at all the potential target cells at all times in order to minimize the data forwarding delay. Because there is no soft handover support in the original LTE network, soft handovers can be introduced by employing relay stations or other equipment. Therefore, cooperation of BSs and relay stations is a hot topic to study into. More specific information is presented in Section 3.2.

3. Handover schemes

Handover scheme refers to systematic and organized plans including network configuration, handover procedure design, and specific handover process management. Since the handover process is managed by handover schemes, many researches focus on handover schemes. Besides, as one of the most important functionalities of a mobile system, the handover procedure should be designed according to the nature of the network architecture. For example, in a distributed LTE system, the user context is relocated between the eNBs during each handover, while in a centralized GSM system, the user context must route to the MSC. Therefore, different characterized network architectures determine different handover schemes. Both the network and the handover scheme should be addressed. Recent articles focus on reducing the handover delay of HSR communications. There are mainly two ways to reduce this delay: one is to reduce the handover rate, and the other is to reduce the single handover processing time. Note that the handover delay is not the only handover performance metric. Various metrics are discussed throughout this section. In this section, various network access schemes and handover schemes from time-honored to new-styled are discussed and classified according to their characteristics (Fig. 4).

3.1. Main access technologies

3.1.1. Satellite to ground related

Satellite communications have a large coverage, so there is no need to install a lot of ground communication equipment next to the tracks. The Doppler frequency shift almost has no effect on satellite communications. Since the height of the satellite makes the coverage very broad, almost no handover is needed. However, blind spots exist in its coverage area where there are tall city buildings, rugged mountains, and tunnels. In addition, satellite communication links with limited bandwidth (typically 4 MB) and high costs (e.g. satellite renting cost, user equipment cost, etc.) cannot meet the communication requirements of large numbers of passengers. Furthermore, the satellite communication has considerably large link delay (about 500 ms) and long-range multi-trip propagation delay (about 250 ms), making it not suitable for real-time multimedia applications. Besides, the satellite signals suffer high losses in bad weather conditions. The transmitters

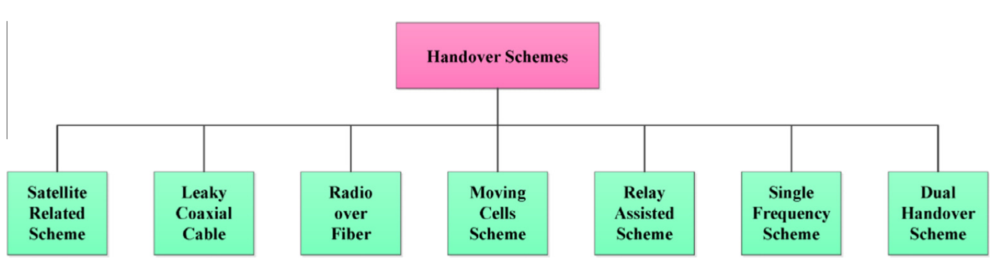

Fig. 4. Classification of handover schemes.

and receivers mounted on the top of the HST may have an effect on the safety of train operation.

The original Universal Mobile Telecommunications System (UMTS) and Worldwide Interoperability for Microwave Access (WiMAX) standards did not consider the chain coverage and high-speed mobile scenarios. After applying them to the HSR communications, many problems ensued, e.g. the sharp decline of received signals caused by the Doppler frequency shift and the increasing load on the system resource management caused by fast and frequent handovers. A network model with Remote Radio Unit (RRU) and Building Baseband Unit (BBU) can expand cell coverage to 40–70 km [25]. This coverage expansion reduces the handover rate significantly. However, this scheme still needs a large number of RRUs along the railway. Due to the very large train running space, these devices are idle when there is no train located within their coverage. Thus, this scheme has a great resource and energy waste. The authors in [26] propose a broadband wireless coverage based on WiMAX for HSR communications. However, the experiments show that the data rate declines sharply during handovers.

Similar to the two-hop architecture, Wireless Local Area Networks (WLAN) are used as complements for loopholes of the satellite communication in [27,28] (Fig. 5). In addition, when the HST stops at a train station, WLAN are used to provide broadband Internet access and update video resource for the in-train multimedia system. If we only use WLAN to cover the high-speed rail, the deployment of many WLAN nodes will lead to engineering complexity and high construction costs. The authors in [29] propose a communication system which uses Wi-Fi to provide links between the Internet and high-speed train passengers. In this system, layer 2 (L2) and layer 3 (L3, i.e. Mobile-IP) handovers are considered. The results show that the proposed system realizes a maximum of 25 Mbps (an average of 16 Mbps) of UDP (User Datagram Protocol) throughput and an average of 110 ms of L2HO time, when the HST is traveling at 270 km/h.

Considering the advantages and disadvantages of the satellite coverage and the WiMAX coverage, a combination of both schemes is a way out. The authors in [30] combine Global Positioning System (GPS) technology and 802.16e technology to achieve fast handovers in the railway communication system. Depending on the GPS information, a candidate base station is selected as the target BS. The result reveals that compared with conventional schemes, the proposed scheme can save the handover operation time of about 180–420 ms.

3.1.2. Leaky coaxial cable

Leaky coaxial cable (LCX) scheme provides a uniform coverage of radio signals without mutual interference. It reuses several frequency points in the long segment of the HSR to achieve high bandwidth efficiency. This broadband system has high data rates, strong adaptability to the terrain, and many other advantages of high-capacity digital communications. In [31], the authors made an experiment of LCX (Fig. 6) in the 2.4 GHz band for train-ground information transmission. The results show that the transmitting rate in LCX can be 20 Mbps and is more stable than that in free space.

However, due to heavy radio signal losses in LCX, many repeaters need to be installed. And the stringent requirements on the slot size make the cable cost very high. These two reasons increase the relay coverage cost substantially. In addition, the LCX transmission indicators are also quite demanding. The transceiver and repeater equipment of LCX are pretty intricate. Recent researches mainly focus on the cable-making crafts. In a nutshell, its networking initial investment cost would be very high. So the LCX network coverage solution is not suitable for the entire line of the HSR. Instead, it is an alternative to other wireless access methods in specific conditions such as tunnels.

3.1.3. Radio over fiber

In the future cellular mobile communication systems, user demands for broadband communications are increasing. In order to keep up with this pace, reducing the cell radius to increase the system capacity and spectral efficiency is a solution. However, smaller cell size will make handovers become more frequent. Therefore, the acceptable handover latency becomes less. So, the performance of fast handovers is particularly important in this trend. A recently novel framework called radio over fiber (RoF) [32–37] which shows a promising future is very suitable for the HSR broadband wireless access system. Distributed antenna systems (DAS) in conjunction with RoF technologies have been applied to the HSR communications. Multiple RAUs are located along the railway and connected to the control center via a fiber ring. The electrical signals at the control center are converted into optical signals and transmitted to the RAUs via the fiber ring. Then they are converted back into electrical signals and radiated by the antennas from RAUs (Fig. 7). The 60 GHz technology is applied to gain a much broader bandwidth (e.g. 1 GB) compared with current access technologies. The RoF scheme is cost effective since one

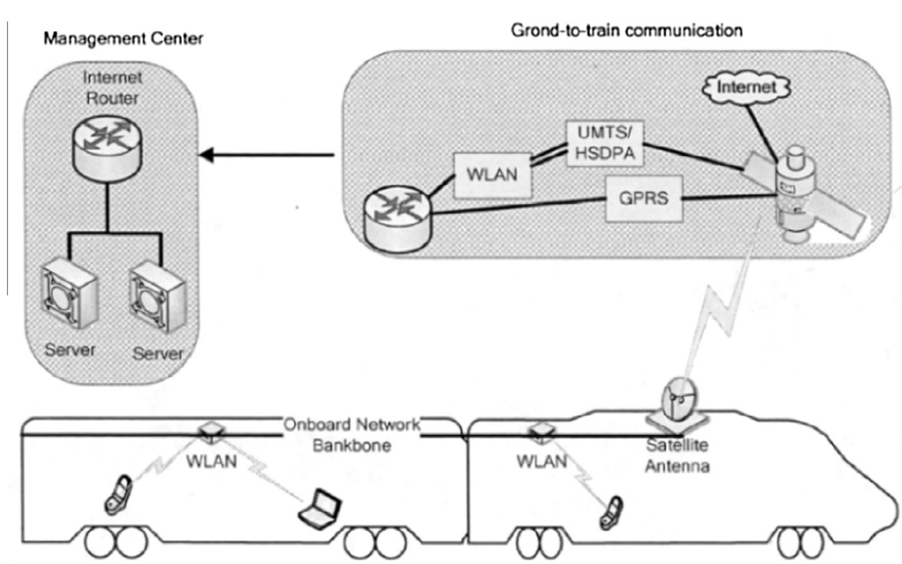

Fig. 5. Satellite combined with WLAN.

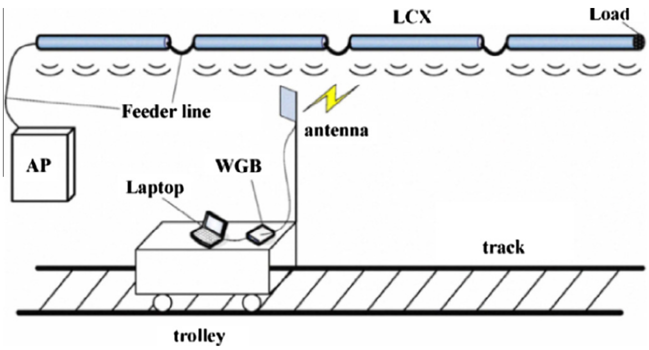

Fig. 6. LCX in field test.

control center can be associated with many RAUs. These micro-cells which usually have a radius of 100 m are equivalent to each other. The radiated area of micro-cell antennas can be linear rather than omnidirectional. Moreover, when the HST moves between RAUs under the same control center, no handover is needed. Thus, compared with conventional schemes, the RoF scheme can significantly reduce the handover rate. Besides, the handover time can be reduced since the optical switching is much faster (e.g. optical switching time is around the order of μs) than the electrical switching.

RoF has a variety of handover strategies. The authors in [38] propose a new type of cell arranging strategy, namely Group Cell, in the RoF distributed system. The group cell refers to a group of geographically adjacent cells. These cells use the same set of communication resources (time slots, frequencies, etc.) at the same time on the same user terminal and use different sets of communication resources on different terminals. Based on the group cell structure, a new handover scheme called Group Handover, i.e. handovers only occur between group cells, arises. When the user is moving within the group cell, no handover is needed. The results show that the call dropping probability is negligible and the bandwidth utilization is high. However, similar to the traditional cellular handovers, group handover does not suit for the RoF scheme in high-speed railway scenarios. Further improvements should be made to this handover strategy.

3.2. Novel handover schemes

3.2.1. Moving cells

The concept of moving cells (MC) is initially proposed for the highway communications [8]. The base stations are physically moving on the tracks beside the highways. So the cell formed by a base station is also actually moving. However, the construction of many actually moving base stations is not practical. In fact, it sparked some innovative ideas for realizing the function of moving cells.

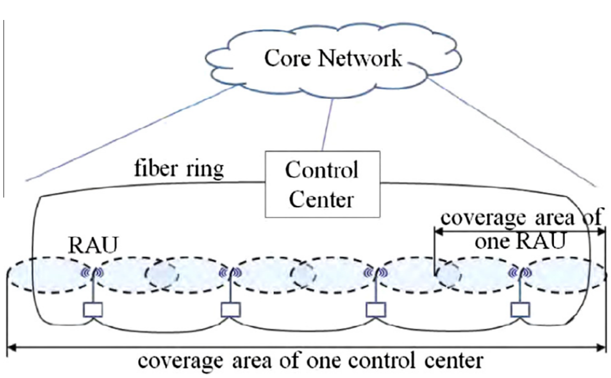

Fig. 7. Linear cell planning based on RoF.

In the RoF system, each BS is no longer associated with one fixed RAU. In addition, the number of BSs can be far less than the number of RAUs. Furthermore, if the RAUs use the same frequency, it is also possible to use the same radio channel between the RAUs and the HST. To this end, there is no need to create a new low-layer connection between RAUs and the HST. The only thing that the BS needs to do is to transfer data to another RAU. Therefore, no synchronization or conversion of radio channels is required between the BSs. Even if the RAUs use different frequencies, handovers can still be avoided among RAUs controlled by the same BS or control center through the "moving frequency" concept [33]. The basic idea of the moving frequency is to change the radio frequencies of the RAUs alternatively in a predefined order according to the location of the HST. As a result, the radio frequency used by the HST will be unchanged and the frequencies of RAUs are "moving" with the HST. However, when the HST moves into the coverage area of another BS or control center, the handover is still needed. Needless to say, the handover rate is reduced to a considerably low level by this scheme.

In [35], the authors propose a "moving extended cells" concept. The main idea is to introduce a user-centric virtual extended cell which includes the source cell and surrounding cells of the source cell. Same frequency is used to transmit the same data to the same user. A restructuring mechanism is applied to the virtual extended cell according to the user's mobility. In this way, the virtual cell group always makes the user in its center and moves together with the user. Multicasting and 60 GHz techniques are also used in this scheme to provide seamlessly handovers. Simulation results show that it can support 100 Mb/s data rate when the moving speeds are less than 144 km/h. However, this cell configuration takes up a lot of resources and generates a lot of interference (e.g. redundant data transmissions). Its transmission distance is limited by the millimeter technology. Therefore, this scheme needs further improvements in order to support train communications at higher speeds.

Another variation of moving cells is proposed in [39]. A Cell Array smartly organizes a number of cells (e.g. three cells in the article) along the railway (Fig. 8). Specifically, when the HST is in Cell A, the Cell A, B, and C form Cell Array I. If the HST has entered Cell B completely, Cell A will be removed from Cell Array I, and Cell D is added in. Then Cell B, C, and D form Cell Array II. In the interior of the HST, there are several "moving" LTE femtocells which are equivalent to eNBs. The femtocell aggregates service traffic within one train carriage (c.f. the two-hop architecture). Since the moving direction and speed information of the HST is generally known, the Cell Array predicts the target LTE cell to achieve seamless handovers. Scheduling and resource allocation mechanisms are also considered (see Section 4.2.2). The results show that lower handover latency (about 5–18%) and higher data throughput (above 150 Mbps) are obtained compared with normal LTE access. In addition, it also well resists to network and traffic dynamics. Thus, it can provide uninterrupted services. We find that this Cell Array actually has only two active cells. So the 3-cell configuration is a bit redundant. However, this redundant configuration has the robustness for network dynamics.

3.2.2. Relay assisted handover

Relay station (RS) is widely used in communication systems in order to increase capacity, expand network coverage, and enhance weak-field zone. The RS can even have a signal processing function. High-speed mobility will bring about problems like frequent handovers, severe Doppler Effects, and the postponement of handover triggering location. Therefore, RSs can be employed to improve the handover performances.

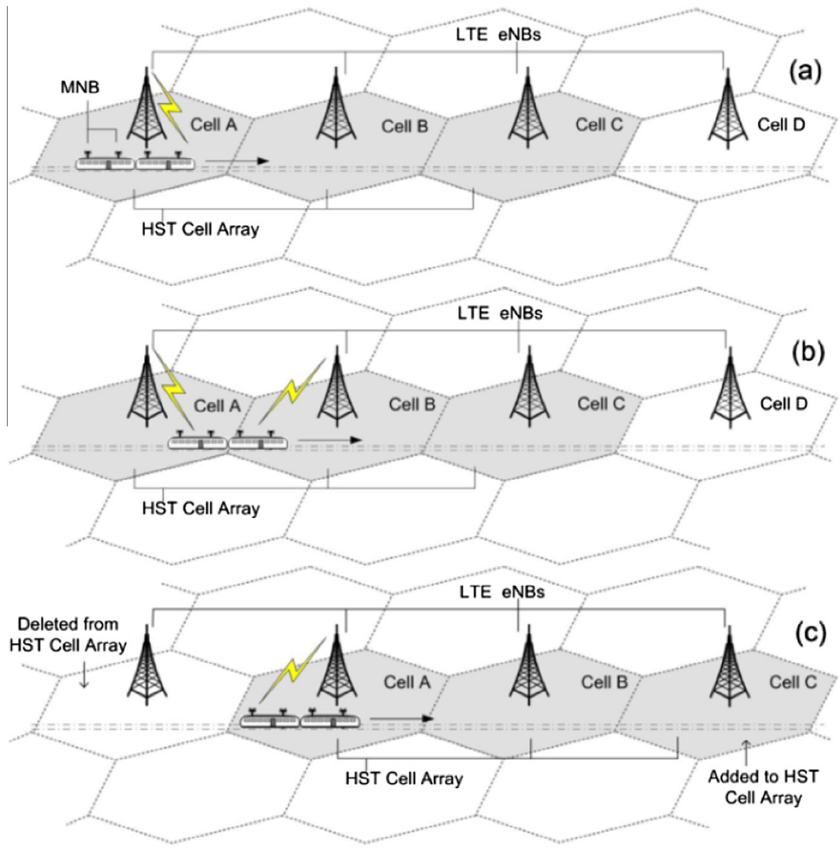

Fig. 8. Moving Cell Array.

The authors in [40] introduce a relay assisted power control scheme which ensures the signal coverage and strength to meet the minimum reliable communication requirements in the overlapping area. In this scheme (Fig. 9), the RS is located in the middle of the overlapping region and is only associated with the source BS. Besides, the speed and positioning information provided by the train control system is used to assist handovers. The RS first acts as a repeater (i.e. the HST is at d_1), so that the diversity gain is provided to the HSR communication. When the handover triggering condition is satisfied, the RS starts the power control (i.e. the HST is at d_2) to maintain the communication link between the HST and the source BS, while the target BS is preparing for the handover. After the handover is completed, the RS stops the transmission. In fact, this scheme pushes back the handover trigger time in order to avoid premature triggering which may lead to communication interruptions and handover failures. This scheme improves the handover success probability, because the coverage area of the source BS is expanded by the power control of the RS. However, additional relay nodes complicate the system interference analysis. Besides, the mentioned article does not consider the Doppler Effect and the multipath effect. We think these effects will greatly affect the relay handover performances.

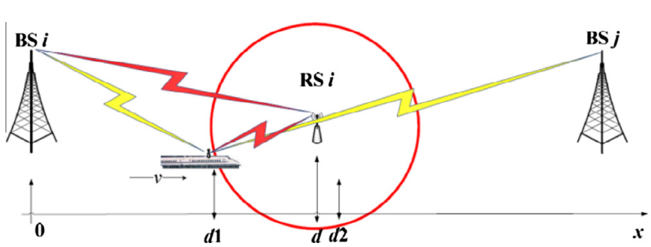

Fig. 9. Relay assisted handover.

Network coding [41] is applied in some relay assisted handover schemes. For example, in [42], each relay station can transmit a combined signaling (i.e. handover control data) after encoding the two signaling from two adjacent cells. In this way, each MS receives data from the source BS and the RS in the cell boundary, respectively. Therefore, macro diversity is gained in this scheme. To avoid interference between the BSs, the system uses time division or frequency division pattern to send data. The results show that the outage probability can be reduced to about 10.5%, which enlarges the cell coverage. However, the encoding procedure must consume a certain period of time. In the time division pattern, this procedure occupies two time slots, while it takes up three time slots without using the network coding. However, the time division pattern cannot be used in HST since it consumes too much time in coding, forwarding, and decoding the data. For some reason, the frequency division pattern is not addressed in the mentioned paper.

The authors in [43] modified the standard LTE handover scheme via applying a mobile relay station mounted on the top of the HST. In this scheme, two reference points are introduced: Reference Trigger Point (RTP) and Reference Handover Point (RHP). RTP is the point that the measurement result starts to report periodically. RHP is the point that the handover execution process starts. Pre-preparation and bi-casting (see Section 3.2.4) are also introduced to realize seamless handovers. The results show that when the train velocity is higher than 60 m/s, the handover delay reduces to about 50 ms, the bi-casting time reduces to about 1 s, and the communication interruption time reduces by about 27%. Note that the proposed architecture is very similar to the two-hop architecture.

3.2.3. Single frequency handover

It is reasonable to suppose that all the cells in a cellular system of the HSR operate at the same frequency, just like the single frequency network (SFN). Since the whole HST can be treated as a single user terminal (c.f. the first hop in the two-hop architecture), the handover processes can be simplified. Further assuming that all the BSs in the SFN are synchronized, there is no need to establish new low-layer connections. However, when there is more than one HST in the system, different orthogonal resources should be allocated to each HST. Two types of orthogonal resources are commonly used: (1) Code Division Multiple Access (CDMA) [21,22,44], i.e. different code channels can be allocated to different HSTs in order to avoid strong multi-access interferences. (2) Frequency Division Multiple Access (FDMA), i.e. different frequencies are used by different HSTs. Note that the Time Division Multiple Access (TDMA) is not considered, because the synchronizing and the partial-time transmitting characteristics are not suitable for the HSR communication scenarios.

An FDMA scheme in conjunction with the Coordinated Multipoint Transmission/Reception (CoMP) technique is used for the HSR communications in [45]. The CoMP technique helps to achieve soft handovers in the LTE system. This scheme divides the total bandwidth into two equal-sized portions. One portion of the bandwidth is assigned to the HSTs moving in the same direction (there are usually two directions). Two adjacent cells form a Cooperative Transmission Set (CTS). Two eNBs in the same CTS work cooperatively. Same frequency is used to send data to an HST within the CTS. Thus, the soft handover is available at the cell boundary (Fig. 10). The CTS continuously reconstructs along with the running direction of the HST. From an overall point of view, only one portion of the bandwidth is used by one HST during the whole train operation. Therefore, each HST can achieve diversity gain and seamless handovers. The results show that this scheme can enhance the system capacity, improve the bandwidth efficiency, reduce the outage probability to around 1%, and also increase handover success probability to around 99%.

3.2.4. Dual handover

In wireless communications, if multiple antennas are employed, the Multiple Input Multiple Output (MIMO) model naturally arises. The MIMO technology exploits multipath propagation phenomena rather than eliminates the multipath effects to increase throughput and reduce BER. The authors in [46–49] propose dual-link handover schemes based on the two-hop architecture. In these schemes, two antennas (or relay stations) are installed on the front end and the rear end of the HST. When the HST is at the cell boundary, the front antenna (or relay station) performs handover to the target BS, while the rear antenna (or relay station) is still communicating with the source BS. In this way, the network connection can be maintained during the entire handover procedure. The results show that with the help of dual antennas, soft handovers can be effectively seamless, the handover interruption time is reduced, higher handover probability and handover success probability are guaranteed, and the system throughput is also improved.

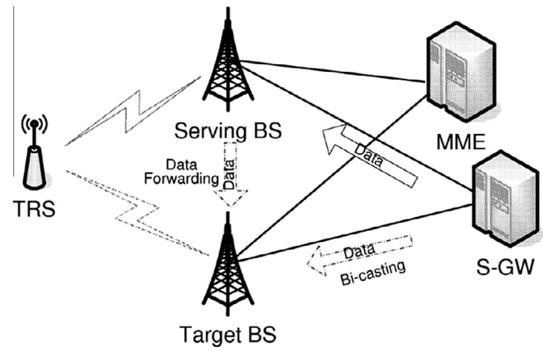

Fig. 11. Bi-casting with handover.

Bi-casting is a basic form of data multi-casting schemes which are used to support fast handovers in the standard LTE system. The goal of this technique is to eliminate the data forwarding delay between the source BS and the target BS (Fig. 11). Based on the radio signal measurements, a dynamic candidate cell set is maintained for a user equipment (UE). If the signal strength of a cell is above a certain threshold, this cell will be added to the candidate set. Then the service data will be multi-casted to all the cells that belong to this candidate set. This scheme makes the data available at all potential target cells before handovers occur. In other words, this method is similar to pre-caching strategy. However, multi-casting and caching data to all these cells consumes numerous backhaul and buffer resources. So the question is, "when is the best time to start the data casting?" According to the linear cell coverage in the HSR scenario, a candidate cell set that contains two cells is enough. So bi-casting is a research hot point in HSR communications. The authors in [50] propose a scheme called reactive data bi-casting which makes some modifications to the 3GPP bi-casting procedures. The main idea is that it triggers data bi-casting after the handover is actually initiated. In this way, it can significantly mitigate the backhaul and buffer resource requirements. In [51], an improved bi-casting scheme based on the speed of the HST is proposed. It estimates the appropriate bi-casting starting time in order to pre-arrange backbone resources and improve backbone efficiency. The simulation results show that bi-casting time is reduced by about 34% in the vehicular scenario, so that the real-time performance is further improved. However, it also reveals the higher the speed, the smaller the improvement. Application of this scheme in a HSR scenario needs to change the composition of the candidate cell set according to the linear cell coverage, e.g. a candidate cell set that contains two adjacent cells.

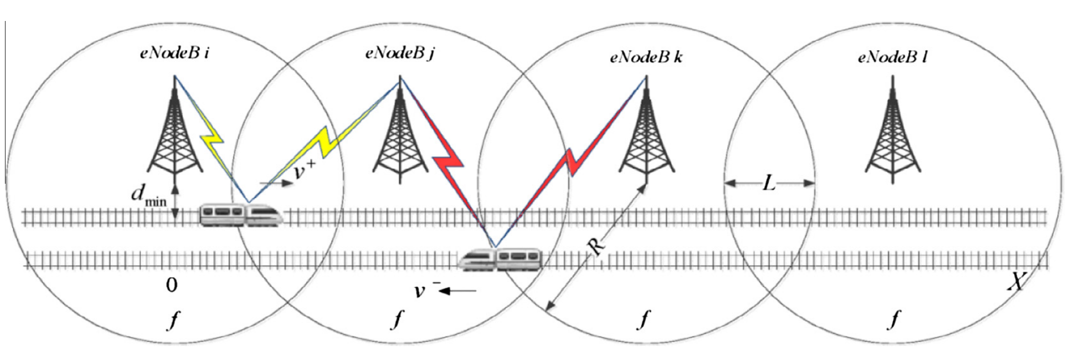

Fig. 10. Co-channel CoMP soft handover.

Related dual-antenna schemes mentioned in this section are summarized in Table 1. Note that bi-casting definitely takes up backhaul resources. However, it can reduce the data transmission time and increase the reliability of handovers. In addition, dual antenna and bi-casting technologies are always combined together. Thus, their advantages can be exploited jointly. In this kind of schemes, the RSs on the HST and the BSs will form a MIMO system. Furthermore, both RSs and BSs can still have multiple antennas. Therefore, this system can benefit from diversity gain, multiplex gain, and seamless soft handovers.

3.3. Performance comparison

In this section, we discuss various handover schemes according to different networks. A performance comparison of network access or handover methods is shown in Table 2. The figures in this table are typical ones under the assumption of a train speed around 350 km/h. It is common knowledge that the interval between two handovers should be far greater than the handover delay, say 100 times larger. And the lower the cost factor is or the higher the handover success probability is, the better the scheme performances are. From this viewpoint, the satellite related schemes do not meet the requirements of broadband high-speed train communications, but considering the handover rate factor, its cost does not seem so inefficient, e.g. no ground station is needed. The existence of Thalys (an international high-speed train operator) in Europe may prove that satellite related schemes are still alive. Similar considerations for the LCX scheme conclude that it is not suitable for the whole railway line of HSR communications. Instead, it is an alternative to other wireless access methods in tunnels or other specific conditions. For example, Japan has applied this scheme to the Shinkansen (New Trunk Line, a network of high-speed railway lines in Japan). Except for these two existing schemes, the other four schemes are still under research and test. It can be seen from the table that RoF and MC schemes have the best performance factors. However, to conclude that they are the most appropriate handover schemes for high-speed train communications, more accurate assessments must be further made. Compared with the LCX scheme, the RS-assisted schemes can improve some performance metrics. However, the cost of establishing many new relay stations is too high. Therefore, it is not appropriate for the coverage of the entire railway line. Instead, it is a better solution in weak fields along the railway lines. Compared with RS-assisted schemes, the SFN and dual handover schemes do not improve much of the listed performance factors. Therefore, improvements and further evaluations should be made before they are chosen to be applied in the high-speed railway communications. After the above analysis and descriptions in Sections 3.1 and 3.2, we argue that emerging schemes like the RoF and moving cells are the most appropriate schemes for high-speed railway communications. The SFN and dual handover schemes also have evaluated good performances. The LCX and RS-assisted schemes should be employed in special situations. The satellite related schemes can still remain existence since direct links can be made to the train. So they could be standby systems in case that the ground systems were destroyed or in case of emergency situations. Since a reliable evaluation needs much more specific and accurate data, our comparisons do have limitations in terms of the data acquisition. With the development of these technologies, the performance metrics of these handover schemes are changing accordingly. Therefore, evaluation methods should change as well. We can conclude that great improvements have been made through the handover researches on communications of high-speed scenarios. However, in order to achieve better performance or get accrete real data for evaluations, we suggest that great efforts should be made into the last four schemes.

4. Handover algorithm

Handover algorithm refers to a step-by-step procedure for solving a problem of handover decision criteria or handover resource allocation. This procedure is mainly done by a computer processor and is often evaluated by metrics such as complexity, correctness, and robustness. The quality of an algorithm is critical to the efficiency of a program and even the whole system.

Handover failure will result in termination of an ongoing call. The forced call termination is more annoying than the call blocks. Therefore, from the user's point of view, the service of a handover request is more important than the service of a new call request. In order to support the user QoS and to provide ubiquitous coverage,

Table 1
Comparison of the dual handover schemes.

Articles	Main ideas	Efficiency description	Reliability description
[46]	Soft handover	Throughput increases by 50% Handover probability 98.7%	Handover success probability 99.8% Outage probability < 2.5%
[49]	Soft handover, modified frame structure	Interruption time reduced by 50%	Call dropping rate < 1%
[47,48]	Soft handover, bi-casting	Throughput increases by 10% Handover probability 99.2%	Handover success probability 99.9% Interruption probability < 10%
[52]	Bi-casting	/	Outage probability < 10% Interruption probability < 20%
[50]	Reactive bi-casting, buffering	Small buffer required Interruption time further reduced	5 packets loss at most
[51]	Velocity-based bi-casting	Resource utilization increased Bi-casting time reduced 34%	/

Table 2
Performance comparison of handover schemes.

Scheme types	Articles	Main techniques	Handover interval (sec.)	Handover delay (sec.)	Handover success probability	Cost
Satellite related	[26–28,30]	TDMA, WLAN, etc.	1 000–10 000	4	0.95	High
LCX	[31]	Leaky feeder, WLAN	1–5	0.1	0.95	High
RoF and MC	[32–39,53]	Optical switch, 60 GHz	50–500	0.005	>0.99	Low
Relay-assisted	[40,42,43,52,54]	GPS, amplify & forward	10–50	0.1	0.95	Middle
SFN	[21,22,44,45]	FDMA, CoMP	30–50	0.5	0.97	Middle
Dual handover	[46–48,50,51]	Bi-casting, soft handover	10–20	0.4	>0.99	Middle

the handover procedures should be further discussed. Current studies on handover algorithms of high-speed mobile scenarios mainly focus on the handover decision criteria and the handover resource allocations. These algorithms are discussed and classified respectively in this section (Fig. 12).

4.1. Decision criteria parameters

Handover decision criteria vary according to different application circumstances. Handover decisions mainly depend on parameters such as threshold level (T), hysteresis margin (h), and timer. Fig. 13 shows some handover parameters when the MS is moving from BS1 to BS2 [55]. The mean signal strength of BS1 decreases as the MS moves away, and the mean signal strength of BS2 increases as the MS approaches. The signal strengths of the two BSs are equal at point A. Four types of handover initiation criteria are introduced as follows. (1) Relative Signal Strength method always lets the MS select the strongest BS. So the handover occurs at position A. However, this method is prone to trigger too many unnecessary handovers, even when the signal strength of the source BS is still at an acceptable level. (2) Relative Signal Strength with Threshold method allows an MS to select a stronger BS when the current signal strength is less than a threshold. For example, the handover occurs at position A if the threshold is T1, at position B if T2, and at position D if T3. In the case of T3, the handover occurs too late that the MS is far into the new cell. Thus, the quality of the communication link from BS1 may be so poor that it causes call drops. In addition, it introduces more interference to co-channel users. (3) Relative Signal Strength with Hysteresis method hands a user over if the target BS is sufficiently stronger than the source BS by a hysteresis margin, say h. In this case, the handover occurs at point C. This scheme also prevents the Ping-Pong effect, i.e. repeated handovers between two BSs caused by rapid fluctuations of the received signal strengths. (4) Relative Signal Strength with Hysteresis and Threshold scheme hands a user over if the current signal level drops below a threshold and the target BS is stronger than the current one by a hysteresis margin. In this situation, the handover occurs at point C if the threshold is T2 and the hysteresis margin is h.

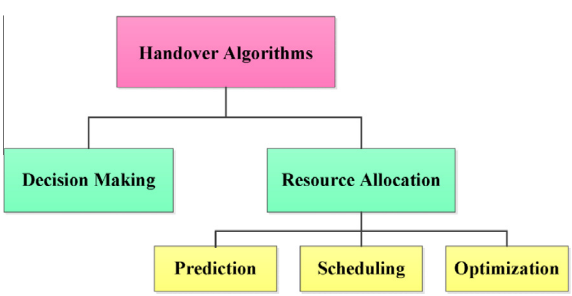

Fig. 12. Classification of handover algorithms.

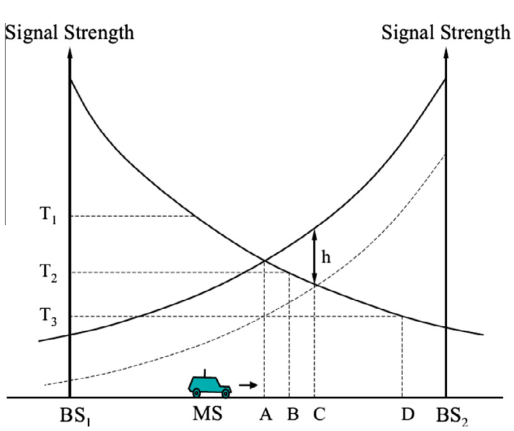

Fig. 13. Handover designs.

According to the above analysis, the adjustment of these handover parameters is quite tricky and is desirable only in some specific circumstances. Therefore, simple adjustments of these parameters cannot improve the overall system performance nor adapt to broad application situations. However, literatures are more likely to engage in these researches. Conventional handover algorithms use averaging methods to reduce fluctuations of the received signal strength at the MS. The authors in [56,57] introduce timers to optimize or modify original averaging methods. The former proposes a timer based algorithm which does not utilize averaging in order to avoid the averaging delay. The results show that the average numbers of handovers (<2) and average numbers of wrong cell handovers (<1) in this algorithm are lower than that in traditional algorithms, respectively. The tradeoff balance between the average number of handover and the crossover point is improved. In addition, this algorithm has the capability to detect the corner effect (i.e. severe degradation of received signal strength due to losing the line of sight path between the mobile terminal and the base station when the mobile is turning around a street corner in a high-density urban area) and is simple to be applied to the receiver. The latter proposes a hard handover algorithm that employs local averaging, a drop timer, and hysteresis. According to the result, no large hysteresis or averaging delay is required, a smaller mean number of handovers is achieved, and a crossover point more than 30 m smaller than that of the traditional hysteresis-based algorithm using exponential averaging can be achieved. The authors also provide a fast and simple numerical procedure to evaluate the performance metrics.

Parameter-optimization handover algorithms in LTE systems mainly aimed at the Time to Trigger (TTT) conditions. In [58], two types of modified handover decision rules based on the UE velocity and the optimized TTT value are proposed. In this algorithm, the hysteresis and TTT parameters are dynamically adjusted according to the measurements of UE velocity and SINR level. The results show that the handover success rate is improved compared with classical A3 event (Neighbor eNB becomes offset better than serving eNB) algorithm in different velocity environments. Besides, when the speed is higher (a speed upper bound exists), the improvement is bigger. The authors in [59] improve the traditional A3 event algorithm to adapt to scenarios that the train speed is less than 350 km/h. This algorithm applies a statistical threshold instead of TTT. The statistical threshold denoted by S is defined as the maximum number of handover criteria that is satisfied. It selects different decision parameter couple (h, S) for different moving speeds, in order to simultaneously reduce Ping-Pong handover (PPHO) and radio link failure (RLF). The PPHO and the RLF are usually a pair of contradictions. The simulation results reveal that several parameter combinations of hysteresis and statistical threshold can be found to maintain both the PPHO probability and the RLF probability below 10%. In [60], the authors take advantage of speed and position information in the overlapping area. The overlapping area is divided into an inner region and an outer region. In this condition, the handover success probability of the inner region should be higher than the outer region. Furthermore, the inner region applies A1 event (Serving becomes better than threshold) algorithm, while the outer region applies A2 event (Serving becomes worse than threshold) algorithm. In other words, system adopts different trigger events according to the positions of the HST. This algorithm can promote handovers to occur at proper places and ensure handovers to trigger at positions that have better signal qualities. Besides, with the increasing of TTT, threshold T is also changing in order to make handovers occur more easily in the inner region. The result shows that the normalized handover success probability is more than 81% when the HST speed is less than 450 km/h. We can conclude from these algorithms that adjusting decision parameters to appropriate values helps to trigger handovers more timely under different scenarios.

4.2. Resource allocation

Resource allocation is used to assign available resources in an economical way and to meet different requirements of different services. The goal of handover resource allocations is to further lower the call dropping probability of handovers. A good handover resource allocation algorithm has two major aspects — efficiency and optimality. The former refers to allocate the handover resource within the shortest time. The latter refers to allocate an appropriate amount handover resource to each user so that the network performance is the best. Prediction methods can reduce the resource allocation time. Scheduling methods are used to deal with users of different priorities. Optimization methods, which calculate the appropriate resources needed by each user considering the user QoS and system resource constraints, will provide better user QoS and optimize the network load. However, some trade-offs have to be made between the allocation time and the optimality of allocation solutions.

4.2.1. Prediction techniques

According to the space–time characteristics of the train operation, the auxiliary information generated during train operations can be used to assist handovers. The assistant information is used to predict the best handover reference point. Prediction techniques make handover decisions based on the expected future values of the received signal strength or other indicators. For example, in terms of unnecessary handover reduction, prediction methods perform better than the Relative Signal Strength method. Thus, the handover processing time will reduce and the handover success probability will increase. Various handover prediction algorithms have been proposed aiming to improve the handover performances.

In [61], an algorithm which is suitable for the high-speed railway environment is proposed. The main idea is that by using the GPS information of train location, speed, and direction, the system firstly completes the channel assignation and activation, and then makes handover decisions according to the measurement report before the train enters the overlapping area. The results show that this algorithm can reduce the inter-MSC handover delay by about 64%. However, a large amount of statistical location information should be acquired in order to get a good estimation of the handover reference point. In addition, handover reference points should be updated periodically. The authors in [53] propose an optical switching scheme based on position-tracking algorithm to realize "moving cells" in a RoF network. This algorithm exploits auxiliary information such as the HST position, speed, and direction, so that the exact time of the HST entering the next RAU is known. Therefore, prediction of the optical switching time is possible. Then the train control center will pre-allocate radio resources and pre-activate corresponding channels in the target cell before the HST enters the overlapping area. The algorithm eliminates the connection signaling of request and response. The simulation results reveal that this scheme reduces the handover latency to around 4 ms, achieves a total upstream traffic of around 2 Mb/s, and improves the channel utilization. However, this algorithm assumes that the HST always does a uniform rectilinear motion. We suggest that further studies should remove these two assumptions, since real train scheduling table and geographical data are available in the train control center. Both of the mentioned algorithms require a plenty of prior information, storage capability, and computation capability. In addition, if the predictions or calculations are inaccurate, there will be a series of serious errors. There still exist blanks in the impact evaluation of inaccurate prediction or calculation. Needless to say, prediction methods really help shorten the total handover time. However, correctness and processing capability are two key aspects in evaluating this type of methods.

4.2.2. Scheduling and optimization

Generally, there are two types of services, i.e. handover services and new services. In the following paragraphs, we denote these two types of services as handover calls and new calls, respectively. Since the wireless resource is limited, an efficient handover algorithm should enable a high utilization of the wireless channel capacity and not destroy the online users' QoS such as delay, new call blocking probability, and handover call dropping probability. Because all the requests of new calls and handover calls cannot be satisfied simultaneously, scheduling and queuing algorithms are required to allocate limited resources to these calls. Call admission control mechanism [62,63] determines whether to admit a call considering the current network available resources. From the user's perspective, the termination of an ongoing call is more annoying than the blocking of a new call. Therefore, some call admission control strategies give handover calls higher priorities over new calls. This kind of scheduling is also called the handover prioritization [16]. The handover prioritization schemes improve the performance of handover calls, however, at the expenses of less admitted traffic and higher new call blocking probability. The authors in [64] provide a predictive channel reservation method for the road vehicle communication system based on RoF. According to the theory of M/M/c queuing model, the network traffic load and the speed of vehicles are adjusted by a numerical analysis method to obtain the best size of the pre-reservation area under different situations. The results show that the handover call dropping probability decreases to about 1.5%, and the new call blocking probability increases to about 3% when the speed of vehicles is 30 m/s. However, this algorithm does not consider the handover delay. The effectiveness and accuracy of the numerical analysis are limited by the assumptions of queuing model and zero handover delay. The authors in [25] propose an effective Radio Admission Control (RAC) algorithm based on a Rician shading model for high-speed railway communications with MIMO antennas. Different service priorities are considered to differentiate simultaneous access requests of multiple users. Simulation results show that the proposed algorithm has higher call blocking and call dropping probabilities than the existing RAC mechanism; however, the QoS of different services can be guaranteed at higher moving speeds.

When designing handover priority policies and queuing schedules, the following methods can be used to firstly guarantee the continuity of real-time services and higher prioritized applications. These methods are sharing method, borrowing method, and subchannel-combine method. In the sharing method, a portion of "free bandwidth" defined as the bandwidth neither allocated to nor reserved for any call is allocated to a new admitted call of any class. This free bandwidth may vary during the service time according to different situations. In [65], the authors allow dynamic bandwidth sharing between different traffic types in order to improve bandwidth utilization in mobile networks. The results demonstrate that the call blocking probabilities of real-time and non-real-time applications are less than 20% and 30%, respectively, and a higher efficiency is achieved. The borrowing method provides a higher priority service through the seizure of a lower priority service's resource in the target cell. The authors in [66] develop a preemptive channel borrowing method to promote the channel utilization.

The total channels are divided into two parts: real-time service channels (RC) and non-real-time service channels (NC). A real-time handover request can borrow a channel from NC, while a non-real-time request (new call or handover call) can only borrow an idle channel from RC. The results show that in terms of call blocking probability, the preemptive borrowing performs around 0.1% and 1% better than the non-preemptive borrowing for real-time and non-real-time service calls, respectively. However, the handover rate of non-real-time services increases. The sub-channel-combine method is applied in a multi-carrier cellular mobile communication system in [67]. This handover algorithm lets the target cell and the source cell assign sub-channels jointly. These sub-channels constitute a combined physical channel group. The MS communicates with the source BS and the target BS at the same time. The numerical analysis shows that both real-time and non-real-time handover call dropping probabilities decrease while the new call blocking probabilities increase. The performances of real-time services are better than that of non-real-time services. In addition, the complexity introduced by providing sub-channels from multiple base stations is similar to the soft handover. Therefore, this method can obtain comparable diversity gain, and there is no redundant resource occupation. Besides, this method can be combined with other resource reservation methods which are introduced below.

In the above methods, resources are allocated after requests have been made. Unlike those methods, dynamic resource reservation (RR) is a kind of pre-allocation method in which resources are reserved for potential handovers before requests have been made. Dynamic resource reservation makes the statistics of the channel occupancy rate and the channel utilization rate at current time segment, and the handover blocking rate at previous time segment in the target cell. Then it predicts the handover blocking rate of the next t time. Based on the results of the dynamic prediction, the channel resources will be reserved for potential handovers. The authors in [68] propose an efficient dynamically adjusted resource reservation policy considering potential handover calls in the neighbor cells based on the SNR and the traffic profile of each mobile. The results show that higher priority classes of calls will receive lower handover call dropping rate (by 20% of each class) due to the resource reservations. Note that the dynamic resource reservation can also borrow the channel resources from a less real-time sensitive service to meet the current urgent handover request. This method reduces the handover blocking probability. However, it does not reduce the handover latency, since the handover procedure is not changed. It is worth noting that the Markov Chain Model acts as a popular mathematical tool for the analysis and evaluation of the above methods. Related researches are relatively rare in high-speed railway scenarios, so we suggest that more efforts should be made to apply these methods to the HSR communications.

Most allocation methods need optimality analysis, especially the convex optimization and network optimization analysis. There are many mathematical tricks that can be employed in the analysis. To find a global optimal solution that optimizes the network resource assignation with many constraints, such as every user's QoS requests (bandwidth, data rate, BER, etc.) and the system load balance, in a certain amount of time (solution time should be much smaller compared with the average handover time), is a challenging task. Generally, the global optimal solution is hard to find, and it always takes a long time to converge to the global optimal solution. Thus, in order to meet the real-time computing request, local optimal solutions or suboptimal solutions can be alternatives without deviating from the global optimal solution performance too much. However, to find a local optimal or suboptimal solution can still be time-consuming. Generally, Integer Programming (IP) is utilized to optimize the utilization of system resources. However, the 0–1 IP is an NP-hard problem [69]. The authors in [39] simply use a suboptimal Linear Programming (LP) allocation problem to approximate the IP allocation problem. After getting the solution which is not feasible in the original IP problem, a convert-back method is used to map the suboptimal solution to a feasible solution in the original IP problem. Although the solution of this algorithm is not globally optimal, simulation results show that it effectively reduces the handover delay of non-real-time data, reduces the handover failure probability, and improves the system throughput. However, this algorithm is not capable of supporting real-time calculation, since obtaining a suboptimal solution is still time-consuming. Anyway, it is very time-saving compared with solving the original problem. The authors in [70] propose a multi-dimensional resource allocation algorithm for high-speed railway downlink MIMO-OFDM system. Sub-carrier, antenna, time slot, and power are jointly considered to formulate a 4-dimensional nonlinear IP problem. The objective is to minimize the total transmitted power under the constraints of each user's QoS requirements. The optimal and approximate solutions are obtained by linearization and quadratic fitting, respectively. Simulation results have proved that: the approximate solution has lower total transmitted power than the optimal solution while its computational complexity is relatively higher. The reason is that the process of linearization, to some extent, can reduce the overall computational complexity. To this end, we can conclude that trade-offs usually have to be made between system performances and computational complexity in these optimization problems.

4.3. Performance comparison

In this section, we discuss various handover algorithms which are mainly classified into two types. Some algorithms modify or optimize the handover parameters of different decision-making criteria, while other algorithms change the handover procedures according to particular resource allocation application scenarios. A performance comparison of the handover algorithms is shown in Table 3. Note that an algorithm is better if metrics such as handover delay, blocking probability, and the complexity, are lower. The handover parameter modification algorithm is the simplest method. However, a great deal of work in analyzing how to adjust handover parameters should be done in advance according to particular scenarios. Generally, the blocking probability performance of these algorithms has a large range of variation. It is difficult to find the optimal parameter that minimizes the mean and the variation of the blocking probability. Handover parameter modification algorithm is often designed after the handover schemes and models have already been applied. It looks more like a trial and error experiment to find the most appropriate parameters for these handover schemes and models. Therefore, these algorithms have less versatility. However, because of its simplicity and practicability, this method is still recommended to be applied to new handover schemes. In terms of the prediction algorithm, they are mostly used in the RoF, MC, and relay-assisted handover schemes. Through the train position, speed, and moving direction information, these prediction algorithms keep the handover delay in a very low level compared with other traditional algorithms. Their blocking probability is also lower compared with parameter modification algorithms. However, the prediction algorithms have a common weak point which is that the auxiliary information has been assumed to be accurate known information. In fact, obtaining this information needs a variety of data acquisition and estimation techniques. So, the application of this kind of algorithm is a little tricky. In addition, the impact analysis of using inaccurate information is not provided. This analysis gap should be filled by future researches. The scheduling and optimization algorithms mainly consider the application priority and the optimal resource allocation. If an optimal solution is obtained, the blocking probability

Table 3
Performance comparison of handover algorithms.

Algorithm types	Articles	Handover delay (sec.)	Blocking probability	Complexity
Decision parameter	[56–60,71,72]	0.1–1	<0.1	Low
Prediction	[53,61]	0.1–0.5	0.01–0.05	Middle
Scheduling & optimization	[25,39,65–67,70]	0.05–1.2	0.001–0.01	High

could be the lowest. However, the handover delay varies according to different computational and mathematical technologies that are used. The main drawback of this kind of algorithm is the need of additional communication resources (i.e. priority identifications and resource reservations) and high computation capability (i.e. hardware and software requirements). From the handover delay and complexity points of view, the prediction algorithms perform better than traditional algorithms, while scheduling and optimization algorithms have different handover delay performance and they have higher complexity compared with the traditional and the prediction algorithms. We can conclude that excellent algorithms still need to be developed, especially targeting novel handover schemes. We suggest that great efforts should be made to deal with impractical assumptions and drawbacks of the last two types of algorithms. Besides, parameter modifications can serve as minor adjustments and theoretical analysis of algorithms that suit for communications in high-speed mobile scenarios. Thus, better performances of these algorithms will be achieved.

5. Summary and outlook

In this paper, a comprehensive overview of handover schemes and algorithms are given for wireless communication systems in high-speed mobile environments. Although broadband wireless communications have been widely used, most of its current technologies are designed for low-speed mobile (i.e. below 150 km/h) environments. In order to overcome the technical challenges such as fast handover, Doppler frequency shift, and penetration loss caused by the high-speed mobility and to provide higher communication QoS for the end users, especially for the high-speed train passengers, system architecture designs, handover procedure designs, radio resource managements, and cross-layer processing algorithms are presented in this paper. The CPS concept draws a rosy picture of the realization feasibility of future broadband communication systems under high-speed mobile scenarios. Therefore, research directions of future handover schemes and algorithms should combine with the concepts and characteristics of CPS. Modified and newly developed schemes and algorithms that satisfy the CPS requirements should always be encouraged and never stop coming forth.

Novel handover mechanisms and new systems should be designed in order to make handovers more efficient and reliable. The RoF system with moving cells is a great idea that transforms physically moving base stations to logically moving cells. It is an excellent combination of the optical switching technology and the wireless technology. Relay stations can help enhance signal strengths and assist handovers. However, too much interference and over-signaling will be introduced among BSs, RSs, and HSTs. Since a RS is under the control of the BS, the BS of single frequency network can be viewed as an upgraded RS. The cooperation among BSs can provide soft handovers. The SFN schemes also have beneficial interference-avoiding methods. At some level, it may outperform relay assisted handover schemes. Satellites can be seen as air relays in the communication system. However, their altitudes are too high to transmit data within an acceptable latency. Therefore, we can lower the altitudes of the air relay nodes to a position where they still have the line-of-sight property. For example, use air balloons or airships to substitute satellites as relays between the ground stations and the HSTs. The air balloon or airship coverage is much cheaper and has much shorter transmitting latency than the traditional satellite coverage. In terms of dual handover, it is not only limited to the two-cell configuration, but also reasonable for the three-cell configuration (i.e. three-casting). Bi-casting or three-casting can also be applied with the moving cells and the SFN schemes. However, the best time to cast the data to potential cells remains debatable. The HST usually employs two mobile relay stations on the top of its body. All the ground base/relay stations and the train mobile relay stations can employ multiple antennas. These antennas can transmit data separately or cooperatively through the MIMO techniques.

The trend of future network integration is obvious. Heterogeneous networks are emerging in the high-speed train communications. Combinations and enhancements of various existing technologies can mutually reinforce their advantages and complement their disadvantages. Technology integration is also a key element in developing and improving new communication methods, e.g. the integration of optical and wireless, MIMO and multi-casting, CDMA and OFDM, etc. Technology enhancements are narrowing down the performance gaps between ideal and reality and balancing trade-offs such as efficiency and reliability, complexity and real-time performance, etc.

In the light of scheme and system overall considerations in high-speed mobile scenarios, designing particular novel handover algorithms for specific networks, optimizing key indicators of mature algorithms, and testing and converting developed algorithms of other science subjects will make handover algorithms more excellent and versatile. Current handover decision-making criteria have made a few changes. Whether we can find new kinds of decision parameters depends on the handover initiation procedures and the nature of wave propagation. Besides, they are also determined by the network architectures and technologies used. Therefore, if some novel network architectures are created, or some new links are connected between the propagation properties and the parameters that are available to evaluate, then some kinds of new decision criteria will be found. In addition, since "handover" is a key word throughout this paper, the resources used for the handover completion should be allocated carefully and efficiently. On the one hand, prediction methods can shorten the handover latency but sacrifice the resource utilization. On the other hand, scheduling and optimization methods can increase the resource utilization but not improve the handover latency. However, if we can apply these two kinds of methods jointly, in some sense, both performances can be improved. For example, through rough predictions or multistep predictions, partial resources can be allocated before handovers and some pre-scheduling or optimizing procedures can heuristically start at the same time. When the handovers really occur, complete allocation procedures and scheduling changes can be made in the due time.

With the rapid development of the Mobile Internet, handover management solutions in next generation mobility architectures and protocols are definitely based on Mobile IP which provides a flat network platform. Future Internet is envisioned to be open, receptive to different approaches, and based on demands. The chosen approaches depend on several factors such as user mobility characteristics, application types, and QoS provisions, etc.

Elevating the support of Mobile Internet architectures and protocols for higher speed mobility is another approach to realize broadband communications in high-speed mobile environment. We hope all these related developments and improvements will serve as applications in the future CPS systems.

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